

Engineering Lab Report

Dynamic ARP Inspection (DAI), DHCP Snooping & Router Configuration

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1. Executive Summary & Topology Overview

This technical report documents the complete implementation, command execution, and verification of a secure enterprise network topology incorporating a Cisco router acting as a DHCP server, multilayer switches running **DHCP Snooping**, and **Dynamic ARP Inspection (DAI)** to mitigate Layer 2 Man-in-the-Middle (MitM) and ARP spoofing attacks.

Topology Specifications

- **Core Router (R1):** Cisco 2911 running IOS 15.x, configured with DHCP pools and IP gateway services (192.168.1.1/24).
- **Distribution / Access Switches (SW1, SW2):** Catalyst switches configured with DHCP Snooping, DAI, and port security.
- **End Devices (PC1, PC2, PC3):** Client workstations dynamically acquiring IP configurations via DHCP.

2. Router R1 Configuration (DHCP Server Setup)

Router R1 was configured to act as the primary DHCP server for the subnet 192.168.1.0/24. As per lab instructions, specific IP addresses were excluded to reserve static allocations for infrastructure components.

R1 CLI Configuration Sequence

```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
R1(config)#ip dhcp pool POOL1
R1(dhcp-config)#network 192.168.1.0 255.255.255.0
R1(dhcp-config)#dns-server 8.8.8.8
% Invalid input detected at '^' marker.
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#domain-name momolab.com
R1(dhcp-config)#default-router 192.168.1.1
R1(dhcp-config)#ex
R1(config)#
```

3. DHCP Snooping Configuration on Switches (SW1 & SW2)

DHCP Snooping acts as a firewall between untrusted hosts and trusted DHCP servers, preventing rogue DHCP server attacks and building the trusted binding database required by DAI.

SW1 DHCP Snooping Implementation

```
SW1#conf t
SW1(config)#ip dhcp snooping
SW1(config)#ip dhcp snooping vlan 1
SW1(config)#int range g0/1 - 2
SW1(config-if-range)#ip dhcp snooping trust
SW1(config-if-range)#no ip dhcp snooping information option
SW1(config-if-range)#ex
SW1(config)#do wr
Building configuration...
[OK]
```

Verification of DHCP Snooping status and port trust states on SW1:

SW1 Verification: show ip dhcp snooping

```
SW1#do show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
1
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface           Trusted      Rate limit (pps)
-----
GigabitEthernet0/2   yes         unlimited
GigabitEthernet0/1   no          unlimited
```

4. Dynamic ARP Inspection (DAI) Implementation

Dynamic ARP Inspection leverages the DHCP snooping binding database to intercept, log, and discard ARP packets with invalid MAC-to-IP address bindings.

Configuration Steps & Additional Validation Checks

On both SW1 and SW2, DAI was enabled for VLAN 1, uplink ports connected to switches/routers were marked as trusted, and advanced validation checks (checking source MAC, destination MAC, and IP addresses) were enforced.

SW1 & SW2 DAI Configuration

```
SW1(config)#ip arp inspection vlan 1
SW1(config)#int range g0/1 - 2
SW1(config-if-range)#ip arp insp trust
SW1(config-if-range)#ex
SW1(config)#ip arp inspection validate ip src-mac dst-mac
SW1(config)#do wr
```

DAI Interface Verification

Executing `show ip arp inspection interfaces` displays the operational trust state and rate limits across all switch ports:

SW1 & SW2 DAI Interface Statuses

```
SW1#sh ip arp insp int
Interface           Trust State      Rate (pps)      Burst Interval
-----
Fa0/1               Untrusted        15              1
Fa0/2               Untrusted        15              1
Fa0/3               Untrusted        15              1
...
Fa0/24              Untrusted        15              1
Gig0/1              Trusted          15              1
Gig0/2              Trusted          15              1
SW1#
```

5. End-Host Verification & Connectivity Testing

Workstation PC1 was verified using the command prompt to confirm successful DHCP lease acquisition from router R1 and robust end-to-end ICMP connectivity across the secure gateway.

PC1 IP Configuration & Ping Results

```

C:\>ipconfig

FastEthernet0 Connection: (default port)
    Connection-specific DNS Suffix...: momolab.com
    Link-local IPv6 Address.....: FE80::201:64FF:FE32:B922
    IPv4 Address.....: 192.168.1.10
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: 192.168.1.1

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=44ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=12ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 44ms, Average = 14ms

```

6. Summary of Security Posture

Security Mechanism	Target Vulnerability	Implementation Status	Verification Metric
DHCP Snooping	Rogue DHCP Servers & Starvation	ACTIVE (VLAN 1)	Ports G0/1-G0/2 Trusted
Dynamic ARP Inspection	ARP Poisoning / MitM	ACTIVE (VLAN 1)	Access ports untrusted (15 pps)
IP/MAC Validation	MAC Spoofing / IP Hijacking	ENFORCED	validate ip src-mac dst-mac